INCH-POUND
MIL-DTL-23053/12C
17 June 1996
SUPERSEDING
MIL-I-23053/12B
01 May 1986

DETAIL SPECIFICATION SHEET

# INSULATION SLEEVING, ELECTRICAL, HEAT SHRINKABLE, POLYTETRAFLUOROETHYLENE

This specification is approved for use by all Departments and Agencies of the Department of Defense.

The requirements for acquiring the sleeving described herein shall consist of this specification sheet and the issue of the following specification listed in that issue of the Department of Defense Index of Specifications and Standards (DODISS) specified in the solicitation: MIL-DTL-23053.

REQUIREMENTS:

Polymer type: The base polymer used in formulating this sleeving shall be polytetrafluorethylene.

Continuous operating temperature range: -67°C (-88°F) to +250°C (+482°F).

Class: The sleeving shall be furnished in the following classes, as specified (see 6.2a):

Class 1 - Thick wall sleeving
Class 2 - Standard wall sleeving
Class 3 - Thin wall sleeving
Class 4 - Very thin wall sleeving
Class 5 - High shrink, thin wall

Color: The sleeving shall be furnished in clear (transparent milk white to light tan) or in the color specified by the acquiring activity. Colors shall conform to the requirements of Class I of MIL-STD-104.

Longitudinal change: ± 20 percent

Military part number: The Military part number shall consist of the basic number of this specification sheet and dash numbers as follows:


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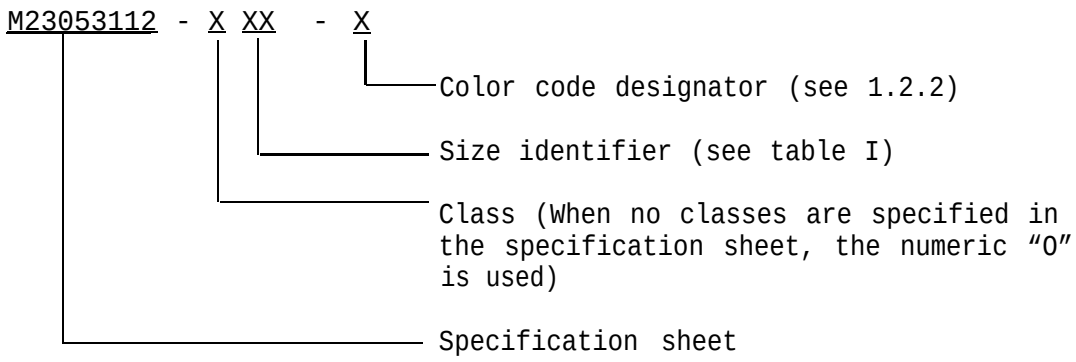


Example: Thin walled (Class 3), 0.430 inch (10.9 mm) as supplied ID, clear sleeving shall be identified as M23053/12-327-C.

TABLE I. Construction details. inches (mm). 1/

| Military part number | As supplied ID minimum | After unrestricted shrinkage | |
|---|---|---|---|
| | | ID maximum | Wall thickness 2/ |
| Class 1 | | | |
| M23053/12-101-* | .166( 4.22) | .130(3.30) | .030 ± .005(.762 ± .127) |
| M23053/12-102-* | .250(6.35) | .193(4.90) | .030 ± .005(.762 ± .127) |
| M23053/12-103-* | .333(8.46) | .257(6.53) | .030 ± .005(.762 ± .127) |
| M23053/12-104-* | .415(10.54) | .320(8.13) | .030 ± .005(.762 ± .127) |
| M23053/12-105-* | .498(12.65) | .383(9.73) | .030 ± .005(.762 ± .127) |
| M23053/12-106-* | .580(14.73) | .448(11.38) | .030 ± .006(.762 ± .152) |
| M23053/12-107-* | .666(16.92) | .510(12.95) | .030 ± .006(.762 ± .152) |
| M23053/12-108-* | .748(19.00) | .572(14.53) | .030 ± .006(.762 ± .152) |
| M23053/12-109-* | .830(21.08) | .637(16.18) | .030 ± .006(.762 ± .152) |
| M23053/12-110-* | .915(23.24) | .700(17.78) | .032 ± .006(.813 ± .152) |
| M23053/12-111-* | 1.000(25.40) | .764(19.41) | .040 ± .007(1.02 ± .178) |
| M23053/12-112-* | 1.170(29.72) | .891(22.63) | .045 ± .007(1.14 ± .178) |
| M23053/12-113-* | 1.330(33.78) | 1.020(25.91) | .050 ± .008(l.27 ± .203) |
| Class 2 | | | |
| M23053/12-201-* | .050(1.27) | .027(.069) | .012 ± .002(.305 ± .051) |
| M23053/12-202-* | .055(1.40) | .032(0.81) | .012 ± .002(.305 ± .051) |
| M23053/12-203-* | .060(1.52) | .039(0.99) | .016 ± .003(.406 ± .076) |
| M23053/12-204-* | .065(1.65) | .043(1.09) | .016 ± .003(.406 ± .076) |
| M23053/12-205-* | .076(1.93) | .049(1.24) | .016 ± .003(.406 ± .076) |
| M23053/12-206-* | .085(2.16) | .054(1.37) | .016 ± .003(.406 ± .076) |



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TABLE I. Construction details, inches (mm). 1/ - Continued

| Military part number | As supplied ID minimum | After unrestricted shrinkage ID maximum | After unrestricted shrinkage Wall thickness 2/ |
|---|---|---|---|
| M23053/12-207-* | .110(2.79) | .067(1.70) | .016 ± .003(.406 ± .076) |
| M23053/12-208-* | .120(3.05) | .072(1.83) | .016 ± .003(.406 ± .076) |
| M23053/12-209-* | .130(3.30) | .080(2.03) | .016 ± .003(.406 ± .076) |
| M23053/12-210-* | .140(3.56) | .080(2.03 | .016 ± .003(.406 ± .076) |
| M23053/12-211-* | .150(3.81) | .089(2.26) | .016 ± .003(.406 ± .076) |
| M23053/12-212-* | .170(4.32) | .101(2.57) | .016 ± .003(.406 ± .076) |
| M23053/12-213-* | .191(4.85) | .112(2.84) | .016 ± .003(.406 ± .076) |
| M23053/12-214-* | .205(5.21) | .124(3.15) | .020 ± .004(.508 ± .102) |
| M23053/12-215-* | .215(5.46) | .130(3.30) | .020 ± .004(.508 ± .102) |
| M23053/12-216-* | .240(6.10) | .141(3.58) | .020 ± .004(.508 ± .102) |
| M23053/12-217-* | .270(6.86) | .158(4.01) | .020 ± .004(.508 ± .102) |
| M23053/12-218-* | .302(7.67) | .178(4.52) | .020 ± .004(.508 ± .102) |
| M23053/12-219-* | .320(8.13) | .198(5.03) | .020 ± .004(.508 ± .102) |
| M23053/12-220-* | .370(9.40) | .224(5.69) | .020 ± .004(.508 ± .102) |
| M23053/12-221-* | .390(9.91) | .249(6.32) | .020 ± .004(.508 ± .102) |
| M23053/12-222-* | .410(10.41) | .260(6.60) | .020 ± .004(.508 ± .102) |
| M23053/12-223-* | .430(10.92) | .278(7.06) | .020 ± .004(.508 ± .102) |
| M23053/12-224-* | .450(11.43) | .311(7.90) | .020 ± .004(.508 ± .102) |
| M23053/12-225-* | .470(11.94) | .329(8.36) | .020 ± .004(.508 ± .102) |
| M23053/12-226-* | .470(11.94) | .347(8.81) | .020 ± .004(.508 ± .102) |
| M23053/12-227-* | .470(11.94) | .334(8.48) | .025 ± .006(.635 ± .152) |
| M23053/12-228-* | .560(14.22) | .399(10.13) | .025 ± .006(.635 ± .152) |
| M23053/12-229-* | .655(16.64) | .462(11.73) | .025 ± .006(.635 ± .152) |
| M23053/12-230-* | .750(19.05) | .524(13.31) | .025 ± .006(.635 ± .152) |
| M23053/12-231-* | .930(23.62) | .655(16.64) | .030 ± .006(.762 ± .152) |
| M23053/12-232-* | 1.125(28.58) | .786(19.96) | .035 ± .008(.889 ± .203) |
| M23053/12-233-* | 1.310(33.27) | .911(23.14) | .035 ± .008(.889 ± .203) |
| M23053/12-234-* | 1.500(38.10) | .036(26.31) | .035 ± .008(.899 ± .203) |
| Class 3 | | | |
| M23053/12-301-* | .034(0.86) | .015(0.38) | .009 ± .002(.229 ± .051) |
| M23053/12-302-* | .038(0.97) | .018(0.46) | .009 ± .002(.229 ± .051) |
| M23053/12-303-* | .046(1.17) | .022(0.56) | .010 ± .003(.254 ± .076) |
| M23053/12-304-* | .050(1.27) | .027(0.69) | .010 ± .002(.254 ± .076) |
| M23053/12-305-* | .055(1.40) | .032(0.81) | .012 ± .003(.305 ± .076) |
| M23053/12-306-* | .060(1.52) | .039(0.99) | .012 ± .003(.305 ± .076) |
| M23053/12-307-* | .065(1.65) | .043(1.09) | .012 ± .003(.305 ± .076) |
| M23053/12-308-* | .076(1.93) | .049(1.24) | .012 ± .003(.305 ± .076) |
| M23053/12-309-* | .085(2.16) | .054(1.37) | .012 ± .003(.305 ± .076) |



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TABLE I. Construction details, inches (mm). 1/ - Continued

| Military part number | As supplied ID minimum | After unrestricted shrinkage | |
|---|---|---|---|
| | | ID maximum | Wall thickness 2/ |
| M23053/12-310-* | .093(2.36) | .061(1.55) | .012 ± .003(.305 ± .076) |
| M23053/12-311-* | .110(2.79) | .067(1.70) | .012 ± .003(.305 ± .076) |
| M23053/12-312-* | .120(3.05) | .072(1.83) | .012 ± .003(.305 ± .076) |
| M23053/12-313-* | .140(3.56) | .080(2.03) | .012 ± .003(.305 ± .076) |
| M23053/12-314-* | .150(3.81) | .089(2.26) | .012 ± .003(.305 ± .076) |
| M23053/12-315-* | .153(3.89) | .096(2.44) | .012 ± .003(.305 ± .076) |
| M23053/12-316-* | .170(4.32) | .101(2.57) | .012 ± .003(.305 ± .076) |
| M23053/12-317-* | .191(4.85) | .112(2.84) | .012 ± .003(.305 ± .076) |
| M23053/12-318-* | .205(5.21) | .124(3.15) | .015 ± .004(.381 ± .102) |
| M23053/12-319-* | .215(5.46) | .130(3.30) | .015 ± .003(.381 ± .076) |
| M23053/12-320-* | .240(6.10) | .141(3.58) | .015 ± .004(.381 ± .102) |
| M23053/12-321-* | .270(6.86) | .158(4.01) | .015 ± .004(.381 ± .102) |
| M23053/12-322-* | .302(7.67) | .178(4.52) | .015 ± .004(.381 ± .102) |
| M23053/12-323-* | .320(8.13) | .198(5.03) | .015 ± .004(.381 ± .102) |
| M23053/12-324-* | .370(9.40) | .224(5.69) | .015 ± .004(.381 ± .102) |
| M23053/12-325-* | .390(9.91) | .249(6.32) | .015 ± .004(.381 ± .102) |
| M23053/12-326-* | .410(10.41 | .260(6.60) | .015 ± .004(.381 ± .102) |
| M23053/12-327-* | .430(10.92 | .278(7.06) | .015 ± .004(.381 ± .102) |
| M23053/12-328-* | .450(11.43 | .311(7.90) | .015 ± .004(.381 ± .102) |
| M23053/12-329-* | .470(11.94 | .329(8.36) | .015 ± .004(.381 ± .102) |
| M23053/12-330-* | .470(11.94 | .347(8.81) | .015 ± .004(.381 ± .102) |
| Class 4 | | | |
| M23053/12-401-* | .050(1.27) | .025(0.64) | .006 ± .002(.152 ± .051) |
| M23053/12-402-* | .055(1.40) | .031(0.79) | .006 ± .002(.152 ± .051) |
| M23053/12-403-* | .060(1.52) | .038(0.97) | .006 ± .002(.152 ± .051) |
| M23053/12-404-* | .065(1.65) | .043(1.09) | .006 ± .002(.152 ± .051) |
| M23053/12-405-* | .076(1.93) | .046(1.17) | .006 ± .002(.152 ± .051) |
| M23053/12-406-* | .085(2.16) | .054(1.37) | .006 ± .002(.152 ± .051) |
| M23053/12-407-* | .093(2.36) | .057(1.45) | .006 ± .002(.152 ± .051) |
| M23053/12-408-* | .110(2.79) | .063(1.60) | .006 ± .002(.152 ± .051) |
| M23053/12-409-* | .120(3.05) | .072(1.83) | .008 ± .002(.203 ± .051) |
| M23053/12-410-* | .140(3.56) | .080(2.03) | .008 ± .002(.203 ± .051) |
| M23053/12-411-* | .150(3.81) | .089(2.26) | .008 ± .002(.203 ± .051) |
| M23053/12-412-* | .170(4.32) | .099(2.51) | .008 ± .002(.203 ± .051) |
| M23053/12-413-* | .191(4.85) | .110(2.79) | .008 ± .002(.203 ± .051) |
| M23053/12-414-* | .205(5.21) | .122(3.10) | .008 ± .002(.203 ± .051) |
| M23053/12-415-* | .215(5.46) | .130(3.30) | .008 ± .002(.203 ± .051) |
| M23053/12-416-* | .240(6.10) | .139(3.53) | .008 ± .002(.203 ± .051) |
| M23053/12-417-* | .270(6.86) | .154(3.91) | .008 ± .002(.203 ± .051) |



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TABLE I. Construction details, inches (mm). 1/ - Continued

| Military part number | As supplied ID minimum | After unrestricted shrinkage | |
|---|---|---|---|
| | | ID maximum | Wall thickness 2/ |
| M23053/12-418-* | .302(7.67) | .172(4.37) | .010 ± .003(.254 ± .076) |
| M23053/12-419-* | .320(8.13) | .192(4.88) | .010 ± .003(.254 ± .076) |
| M23053/12-420-* | .370(9.40) | .214(5.44) | .010 ± .003(.254 ± .076) |
| M23053/12-421-* | .390(9.91) | .241(6.12) | .010 ± .003(.254 ± .076) |
| M23053/12-422-* | .410(10.41) | .260(6.60) | .010 ± .003(.254 ± .076) |
| M23053/12-423-* | .430(10.92) | .270(6.86) | .010 ± .003(.254 ± .076) |
| M23053/12-424-* | .450(11.43) | .301(7.65) | .010 ± .003(.254 ± .076) |
| M23053/12-425-* | .470(11.94) | .329(8.36) | .012 ± .003(.305 ± .076) |
| M23053/12-426-* | .470(11.94) | .347(8.81) | .012 ± .003(.305 ± .076) |
| Class 5 | | | |
| M23053/12-501-* | .078(1.98) | .025(.64) | .009 ± .002(.229 ± .051) |
| M23053/12-502-* | .125(3.18) | .037(.94) | .012 ± .002(.305 ± .051) |
| M23053/12-503-* | .187(4.75) | .050(1.27) | .012 ± .002(.305 ± .051) |
| M23053/12-504-* | .250(6.35) | .063(1.60) | .012 ± .002(.305 ± .051) |
| M23053/12-505-* | .312(7.92) | .078(1.98) | .012 ± .002(.305 ± .051) |
| M23053/12-506-* | .375(9.53) | .096(2.44) | .012 ± .002(.305 ± .051) |
| M23053/12-507-* | .438(11.13) | .112(2.84) | .012 ± .002(.305 ± .051) |
| M23053/12-508-* | .500(12.70) | .144(3.66) | .015 ± .004(.381 ± .102) |
| M23053/12-509-* | .562(14.27) | .155(3.94) | .015 ± .004(.381 ± .102) |
| M23053/12-510-* | .625(15.88) | .178(4.52) | .015 ± .004(.381± .102) |
| M23053/12-511-* | .687(17.45) | .198(5.03) | .015 ± .004(.381 ± .102) |
| M23053/12-512-* | .750(19.05) | .224(5.69) | .015 ± .004(.381 ± .102) |
| M23053/12-513-* | .875(22.23) | .244(6.20) | .015 ± .004(.381 ± .102) |
| M23053/12-514-* | 1.000(25.40) | .278(7.06) | .015 ± .004(.381 ± .102) |
| M23053/12-515-* | 1.250(31.75) | .347(8.81) | .015 ± .004(.381 ± .102) |
| M23053/12-516-* | 1.500(38.10) | .400(10.16) | .015 ± .004(.381 ± .102) |
| M23053/12-517-* | 1.750(44.45) | .450(11.43) | .015 ± .004(.381 ± .102) |
| M23053/12-518-* | 2.000(50.80) | .520(13.21) | .020 ± .005(.508 ± .127) |
| M23053/12-519-* | 2.250(57.15) | .585(14.86) | .020 ± .005(.508 ± .127) |
| M23053/12-520-* | 2.500(63.50) | .650(16.51) | .020 ± .005(.508 ± .127) |
| M23053/12-521-* | 2.750(69.85) | .710(18.03) | .020 ± .005(.508 ± .127) |
| M23053/12-522-* | 3.000(88.90) | .775(19.68) | .020 ± .005(.508 ± .127) |
| M23053/12-523-* | 3.250(82.50) | .835(21.21) | .020 ± .005(.508 ± .127) |
| M23053/12-524-* | 3.500(88.90) | .900(22.86) | .020 ± .005(.508 ± .127) |
| M23053/12-525-* | 3.750(92.95) | .960(24.38) | .020 ± .005(.508 ± .127) |
| M23053/12-526-* | 4.000(101.6) | 1.025(26.03) | .020 ± .005(.508 ± .127 |

1/ Diameter limits for the object to be enclosed shall be as recommended in technical data.

2/ Wall thickness values are less when shrinkage is restricted.



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3/ The asterisk (*) in the part number shall be replaced by color code designators (see 1.2.2).

Unrestricted shrinkage: Test method 4.6.5; 350° ± 4°C (662° ± 8°F) for 10 minutes.

TABLE II. Physical properties. 1/

| Characteristic | Requirement | Test procedure and conditions |
|---|---|---|
| As supplied: | | |
| ID, minimum | Table I | 4.6.3 |
| Heat shock | No cracks, flowing or dripping | 4.6.8 400° ± 5°C (752° ± 9°F) |
| Restricted shrinkage | Pass | 4.6.6.1.2 350° ± 4°C (662° ± 7.2°F), 5 minutes |
| Voltage withstand | Pass | 4.6.6.2 |
| After unrestricted shrinkage: | | |
| ID, maximum | Table I | 4.6.3 |
| Wall thickness | Table I | 4.6.3 |
| Tensile strength, psi (MPa), minimum | 2,500(17.3) | 4.6.13 ASTM D638, 2 inches/minute |
| Ultimate elongation, percent, minimum | 200 | 4.6.13 ASTM D638, 2 inches/minute |
| Dielectric strength, volts/mil (Kv/mm), minimum | 800(31.5) | 4.6.2 ASTM D2671 |



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TABLE II. Physical properties. 1/ - Continued

| Characteristic | Requirement | Test procedure and conditions |
|---|---|---|
| Volume resistivity, Ohms-cm, minimum | $10^{18}$ | 4.6.2 ASTM D876 |
| Water absorption, percent, maximum | 0.01 | 4.6.2 ASTM D570, para. 6.l |
| Specific gravity, maximum | 2.20 | 4.6.2 ASTM D792 |
| Low temperature flexibility | No cracking | 4.6.7.1 -65° ± 2°C (-85° ± 4°F) |
| Heat resistance, properties after: | | 4.6.9 350° ± 4°C (662° ± 8°F), 96 hours |
| Tensile strength, psi (MPa), minimum | 2,500 (17.3) | |
| Ultimate elongation, percent, minimum | 200 | |
| Dielectric strength, volts/roil (Kv/mm), minimum | 800 (31.5) | |
| Flammability | Non-flammable | 2/ |
| Corrosion | Non-corrosive | 2/ |
| Fluid resistance | No change | 2/ |
| Fungus resistance | No growth | 2/ |

1/ Unless otherwise specified, all requirements, conditions and procedures are for all classes.

2/ Because of their relative inertness, MIL-I-23053/12 sleevings are not normally subjected to these requirements. However, the acquiring activity (see 6.2) may, at its option, specify the performance of any or all of these requirements.



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Storage conditions: The supplier shall certify storage at 18° to 35°C (65° to 95°F) for a period of four years. Conformance to 3.5. See 3.5.2 for storage life extension.

Intended use: Heat shrinkable polytetrafluoroethylene sleeving is intended for use in areas where resistance to flame and high temperature are required.

Supersedure data: MIL-I-23053/12 Class 5 sleevings have been dropped from this specification, as they are no longer manufactured.

Custodians:
Army - ER
Navy - AS
Air Force - 85

Preparing activity:
Navy - AS

(Project 5970-1129-12)

Review activities:
Army - MI
Navy - EC
Air Force - 99
DLA - GS



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